

Umbo Computer Vision: Delivering deep learning- powered security video streaming with Google Cloud

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ABOUT UMBO



About Umbo Computer Vision

Umbo Computer
entrusts Google (to
run containerized
applications in a
scalable environment
ease of operation
controlling costs

Founded in 2014 and with offices in Taiwan, San Francisco, and London, Umbo Computer Vision is an artificial intelligence company building an autonomous video security platform for communities, organizations, and businesses alike.

Industries: Manufacturing

deep learning workloads on GPUs.

Location: Taiwan

(<https://cloud.google.com/pubsub/>), [Cloud](#)

Google Cloud results

- Doubles speed of development and deployment cycles
- Seamlessly processes up to 5,000 video streams at once
- Supports expansion into new markets

Stopping unauthorized access to land, buildings, and sensitive rooms or zones is a key priority for businesses and

Runs deep learning workloads on graph processing (GPUs) to lower cost on other services

iKala

About iKala

iKala, a Google Cloud Partner, provides AI-driven marketing technology solutions and Google Cloud infrastructure services. With over 40 data scientists and cloud technical specialists on its staff, it helps over 300 enterprises in the Asia Pacific region use Google Cloud to undergo digital transformation and develop applications for artificial intelligence. Its client base covers multiple industries, including ecommerce, media, finance, gaming, and digital advertising.

over 40 data scientists and cloud technical
Google Cloud (<https://cloud.google.com/>) 300

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

Cloud Functions
(<https://cloud.google.com/functions/>)

covers multiple industries, including

Cloud Load Balancing
(<https://cloud.google.com/load-balancing/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Cloud Storage (<https://cloud.google.com/storage/>)

government
organizations.

An effective
video security
system can
help prevent
disruption
caused by
vandalism,
theft, or other
damaging
activities.

However,
according to
Taiwan-
headquartered
smart video
security system
provider Umbo

Computer

Vision

([https://umbocv.
ai/](https://umbocv.ai/))

, many
businesses and
organizations
rely on systems
that generate
large numbers
of false alarms
(for example, a
falling tree
branch may
trigger a
system that

uses motion
sensors to
detect unusual
movement).
Many systems
also require
near-constant
human
monitoring,
making them
vulnerable to
operators' loss
of
concentration
or momentary
absences.

Founded in 2014, Umbo Computer Vision provides autonomous systems that use AI and deep learning to monitor security video streams for suspicious activity. The business has grown to employ 80+ people and expanded its operations from Taiwan to San Francisco and London.



“Traditional video security systems that rely on motion sensors and other technologies typically generate a number of false alarms that distract and reduce the effectiveness of security personnel,” explains Shao-Heng Tai, AI and Infrastructure Engineer at Umbo Computer Vision. “They may also require one person to monitor up to 16 screens at once, which is difficult to do well.”

Powering security video with AI and deep learning

“We use AI and deep learning to enable security personnel to focus on events that require a physical response, such as an intruder climbing a fence to gain access to a sensitive location,” says Tai.

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*—Shao-Heng Tai, AI &
Infrastructure Engineer,
Umbo Computer Vision*

Umbo Computer Vision reviewed its options and found the cloud provided scalability, stability, and agility not available on local infrastructure. However, four years after launching on a traditional cloud service, the business identified opportunities to improve the scalability, cost effectiveness, and performance of its autonomous video security platform.

“What we wanted from the cloud was a stable, cost-effective infrastructure that could support deep learning, enable us to give our customers ubiquitous access to security video streams, and support our planned expansion into new countries,” says Tai. “Our review found that Google Cloud provided the best containerization technologies in the market, allowing us to run our applications in a scalable,

consistent environment and use [Kubeflow](https://www.kubeflow.org/) (https://www.kubeflow.org/) to simplify deployment of our deep learning workloads.”

Bringing iKala on board

Umbo Computer Vision asked [Google Cloud](https://cloud.google.com/) (https://cloud.google.com/) to help with the migration, and the account team put the video security system provider in touch with Google Cloud Partner [iKala](https://gcp.expert/en/) (https://gcp.expert/en/). The partner dedicated three customer engineers to the project, ran workshops to educate Umbo Computer Vision’s technical team—including how to troubleshoot issues—and provided feedback on the Umbo Computer Vision Google Cloud architecture. “The help iKala provided over two months gave us a deeper understanding of Google Cloud and greater confidence in using services such as Google Kubernetes Engine,” says Tai.

The business completed the migration of its deep learning systems—including the infrastructure supporting customer traffic and the data collection and processing that powers its models—in September 2018. Umbo Computer Vision is using 50 to 100 Docker containers in [Google Kubernetes Engine](#) (/kubernetes-engine) that run 10 applications plus associated systems; [Cloud Functions](#) (/functions) to run and scale its applications in a serverless environment; [Pub/Sub](#) (/pubsub) to provide event ingestion and delivery to support the organization’s data pipelines; [Cloud Storage](#) (/storage) to store and archive data; and [Cloud Load](#)

Balancing (/load-balancing) to distribute traffic to optimize scalability and performance.

“The operational
benefits
provided by
Google Cloud
save us over
60% more than if
we were running
our GPU deep
learning
workloads on
another cloud
service.”

—Shao-Heng Tai, AI &
Infrastructure Engineer,
Umbo Computer Vision

Managing 5,000 video streams

Google Kubernetes Engine provides an optimal environment for Umbo Computer Vision to run its heavily-customized models and provide its researchers fine-grained control over data and code.

The organization is running Google Kubernetes Engine nodes on GPUs that provide the power needed to continuously process up to 5,000 video streams at any one time. "Google Cloud gives us fine control over the scaling and operation of our services so we can limit the cost of running deep learning workloads on GPUs," says Tai. "Because we can fine-tune processing and memory and allocate the correct GPU to each node, we can effectively control the utilization of resources."

"The operational benefits provided by Google Cloud save us over 60% more than if we were running our GPU deep learning workloads on another cloud service," he continues.

"Furthermore, using Google Kubernetes Engine enables us to scale our services up and down in a few minutes using a single command line, while updating a service simply entails changing a few configuration files, with the remaining processes happening automatically."

"Automating our
machine
learning jobs
and running
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training.”

—Ping-Lin Chang, Chief
Technology Officer, Umbo
Computer Vision

Doubles speed of development and deployment cycles

Umbo Computer Vision estimates that running continuous integration and continuous delivery on Google Kubernetes Engine has enabled the business to double the speed of its development and deployment cycles. “Releasing new versions and updates is much quicker than doing so using other cloud services, which is greatly increasing our ability to respond to customer requests and launch new products into the market,” says Tai.

“Automating our machine learning jobs and running learning workloads becomes easier and more cost-efficient with Google Cloud's tools and training,” adds Ping-Lin Chang, Chief Technology Officer, Umbo Computer Vision.

The business is now considering using the Istio open source service mesh to improve the visibility of

its applications and manage the authentication, authorization, and encryption of communication between its various microservices.

Powered by the performance, flexibility, and agility of Google Cloud, Umbo Computer Vision is well equipped to realize its ambitions to expand into new markets. “We are preparing to launch a range of new products for our customers and are using a lot of serverless and containerization technologies on Google Cloud to develop and deliver these,” says Tai. “We’re extremely excited about what the future holds for our business with Google Cloud!”

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Reach out to
our team to
see how
Google Cloud
can help your

business.